

Riko Miyagi

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EDUCATION

University of Waterloo: Bachelor of Environmental Studies, Geospatial Data Science (Co-op)

TECHNICAL SKILLS

Programing/Data Analytics Tools: Excel, SQL, R, Tableau, Python, DrRacket, R, Apps Script

GIS Software/Tools: ArcGIS Pro, QGIS, Catalyst, SNAP(Sentinel Application Platform), TSI, Auto Level

EXPERIENCE

Energy Coordinator - Ecobee, Toronto On Sep - Dec 2026
- Incoming intern for fall term

Energy Reporting and Measurement/Verification Assistant - Parity Inc, Toronto On Jan-Apr 2026
- Measured and verified energy and emissions savings from AI-driven HVAC optimization across 200+ commercial buildings in North America by collecting and analyzing building data.
- Produced energy savings reports including baseline comparisons and performance analyses to support guaranteed savings commitments for client portfolios.
- Detected and resolved anomalies in utility billing data across client portfolios; built automation tools in using AppScripts to streamline auditing workflows, reducing manual review time.

Assistant Coach - KW Meibukan Karate Dojo, Waterloo, ON 2019 - Present
- Assisted in teaching children's karate classes, guiding students through drills and skill development.
- Substitute instructor for adult classes of 30+ students, delivering structured lessons independently.

Teaching Assistant - St. Louis summer Japanese language program July - Sept 2022
- Assisted lead instructor in teaching a class of 30+ students, supporting lessons and classroom activities.
- Provided one-on-one help to students and assisted in creating teaching material

PROJECTS

[Heat Pump vs. Gas Furnace Feasibility Study](#) Feb 2026
- Developed a Python-based financial and environmental analysis tool comparing the 15-year total cost of ownership between air-source heat pumps and natural gas furnaces for Ontario homes.
- Modeled energy consumption using Ontario energy rates, and government rebates, finding a \$9,500 cost difference and 94.5% emissions reduction in favor of heat pumps environmentally.
- Produced visualizations and a professional report and prompted an interactive website showcasing results

[Ontario Electricity Demand Analysis](#) Apr 2026
- Cleaned and merged 12 months of IESO smart meter data in R to compare Residential and Small General Service (<50kW) electricity consumption patterns across Ontario
- Found opposing behavioral profiles due to different seasonal demand (residential winter heating vs. commercial summer cooling), informing targeted demand-response and conservation strategies.

[Site Suitability Analysis for a Small Wind Farm](#) July 2026
- Performed multi-criteria GIS analysis across Huron County, Ontario to identify two optimal wind farm sites using Weighted Overlay, Euclidean Distance modeling, and raster reclassification in ArcGIS Pro
- Delivered a client-facing report to the Ontario Government recommending sites with a combined 16 MW generation capacity, supported by professional map layouts

[Japan Tsunami Evacuation Shelter Coverage Analysis](#) Mar 2026
- Processed 115,446-record GSI national shelter database, filtering to 40,391 tsunami-designated sites using attribute-based selection in QGIS.
- Produced publication-ready choropleth map via graduated symbol classification using Natural Breaks (Jenks).